AWK – Basics

AWK Overview

- Pattern scanning and processing language
- data driven not procedural
  - describe data to work with
  - tell it what to do once matching data is found
- great for dealing with files with data in columns
- we'll focus on gawk the GNU implementation

GAWK Program

- Program has the form

```
BEGIN { ... initialization gawk statements ... }
gawk commands to run on each line of the file
END { ... finalization gawk statements ... }
```

GAWK Program

- Each command has the form
- action = one or more statements enc
- pattern can be regex
- no pattern -> action performed on all lines

```
pattern {action}
```

Fields

- each line made up of fields
- field separator distinguishes fields
  - default = space
  - change value of FS to use other
  - or use -F option to change
- reference field by $# ($0 is entire line)
- NF = number of fields on current line
- NR = record number of current line

Running GAWK

- gawk [options] program [input_files]
- gawk -f program_file [input_files]
- can also create script telling it to run with gawk instead
  - #! /usr/bin/env gawk -f
  - could also use path right to gawk
  - script needs to be executable

AWK Patterns

- Patterns can be regex (/regex/)
  - ~ used for matching regex
  - !~ tests for not matching regex
- Pattern can also compare field or variable to value
  - ==, !=, <, <=, >, >=
- BEGIN and END are special patterns
- nothing for pattern -> applies to all records
- can combine patterns with && (and), || (or)

Variables

- can hold strings / numeric values
- typically initialized in BEGIN
- default = initialized to empty string /0
- standard arithmetic operators available - (increment, decrement, modulo, multiply, etc.)